Einstein-VED Theory: Deterministic and Geometric Universe

"The accuracy of our calculation of radii and masses does not depend on probabilistic hypotheses: it is completely deterministic. The only source of uncertainty is the experimental measurement of the mass of particles."

1. Fundamental premise

- **Spacetime** is the only physical reality.
- **Matter** are confined waves of spacetime.
- **Energy** are free, unconfined waves.
- **Reality is completely deterministic**: there is no ontological probability nor wave function collapse.

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2. Number of confinement waves (n)

- The stability of a wave demands **complete phase closure**:

$$L = n\lambda, \quad n \in \mathbb{Z}^+$$

- **n is exact, not arbitrary**: corresponds to the number of complete waves in the confinement.
- Matter: finite n → generates mass and gravity.
- Free energy: n → ∞ → not confined → does not generate gravity.

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3. Elementary particles

Proton (3D volumetric)

$$R_p = \frac{n\hbar}{m_p c}, \quad n = 4$$

- Exact radius within experimental limits. - n = 4 arises from the minimal closure geometry in 4D.

Electron (2D surface)

$$R_e = \frac{n\hbar}{m_e c}, \quad n_{\min} = 137$$

- n = 137 is **the minimum compatible with confinement outside the nucleus**. - Higher orbitals: n > 137 → consecutive states. - The "fine structure constant" (\alpha^{-1} \approx 137) **arises from geometry**, it is not fundamental.

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4. Mass and VED radius

$$R = \frac{n\hbar}{mc} \quad \Rightarrow \quad m = \frac{n\hbar}{Rc}$$

- Mass **emerges from confinement**, it is not primitive. - Larger n → more mass; larger radius → less mass.

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5. Gravity as an emergent property

- Equating the mass in VED and Schwarzschild:

$$\frac{n\hbar}{R_{\text{VED}}c} = \frac{R_s c^2}{2G} \quad \Rightarrow \quad G = \frac{R_{\text{VED}} R_s c^3}{2n\hbar}$$

5.1 Distinction of radii

- **VED radius (R_\text{VED})**: contains n, determines the mass from the internal confinement of the particle.
- **Schwarzschild radius (R_s)**: contains G, represents the external gravitational geometry.
- **They are not equal**, each one reflects a different physical aspect.

5.2 Constant product

- For fixed n, the **product of the radii is constant**:

$$R_{\text{VED}} \cdot R_s = \frac{2n\hbar G}{c^3} = \text{constant}$$

- This ensures that **macroscopic gravity remains stable**, although individual radii may vary according to geometry or excitation mode.
- The constant G **emerges from the interaction of both radii**, not as an independent parameter.

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6. Holographic interpretation

- The universe can be considered **a 2D projection**.
- Gravity **emerges from the geometry of the surface** and from the number of confined particle waves.
- Unconfined free waves (energy) → n → ∞ → do not produce gravity.

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7. Predictions and consequences

1. Proton: n = 4 → exact radius.
2. Electron: n = 137 → first orbital; n > 137 → successive orbitals.
3. G = G_obs emerges for n = 2.
4. Free energy (n → ∞) → without gravity.
5. (\alpha^{-1} = 137) → geometric consequence, not fundamental.
6. Everything derived from **deterministic geometry and waves**, without probabilistic quantum mechanics.

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8. Conceptual synthesis

- **Matter = confined waves**
- **Energy = free waves**

- **n = integer number of confinement waves**
- **Mass and gravity = emergent properties of n and radii**
- **VED and Schwarzschild radii = distinct functions, their constant product defines G**
- **Physical constants (α, G) = consequence of geometry and holography**, not fundamental parameters.

> This framework demonstrates that a **deterministic and geometric universe** can explain matter, energy, orbitals, and gravity with unprecedented numerical and conceptual coherence. The "fine structure constant" and gravity **are no longer mysterious**, but **inevitable consequences of geometry and the number of waves**. The conceptual revolution of the current paradigm is imminent.

EINSTEIN-VED THEORY: COMPLETE AND RIGOROUS DEVELOPMENT

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1. FUNDAMENTAL PRINCIPLES

1.1 Ontology of spacetime

Spacetime is the only existing physical reality. Coordinates: (T, X, Y, Z) or $3S + T$. It is continuous: $\mathbb{R}^4$.

1.2 Matter and energy as modes of spacetime

There is no separation between "container" (spacetime) and "content" (matter). **Matter** is a confined wave of spacetime, while **energy** is an unconfined wave.

1.3 Principle of geometric confinement

A wave can only be stable if it satisfies the resonance condition:

$$L = n\lambda, \quad n \in \mathbb{Z}^+$$

where: - L = confinement path length - λ = wavelength - n = integer number of complete waves

1.4 Absolute determinism

Reality is 100% deterministic. Quantum probability is epistemic ignorance, not ontological. There is no fundamental wave function collapse.

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2. MATHEMATICAL DEVELOPMENT

2.1 Relativistic basis (natural units $c = 1$)

Lorentz transformations:

$$t' = \gamma(t - vS), \quad S' = \gamma(S - vt), \quad \gamma = \frac{1}{\sqrt{1 - v^2}}$$

Differential form:

$$dt' = \gamma(dt - v\,dS), \quad dS' = \gamma(dS - v\,dt)$$

Pythagorean differential relation (fundamental) that is directly deduced:

$$dt^2 = dS^2 + d\tau^2$$

where $d\tau$ is the proper time of the observed and dt is the observer's time.

2.2 Container (3S + T)

- Differentiable manifold of 4 dimensions: M_container = R^3 × R where 3S are spatial dimensions and T is temporal.
- Temporal cut T(t0) allows observing the geometry of the content at an instant.

2.3 Content: matter and energy

The idea is based on **pure geometry**.

Imagine a **wave in space**: it is not limited, it travels freely through the three spatial dimensions. Nothing confines or attracts it; it is **pure energy**. Gravity does not affect it. We can say, without mistake, that this wave is confined within the limit of an **infinite sphere**.

Thus we define what **energy**, the **photon** is: a free mode of spacetime.

But **matter** is different. Imagine a wave confined within a closed space. For this confinement to exist, the wave must reproduce itself **identical on the contour**: if this does not happen, the wave is not confined.

This confinement space requires an **integer number n of complete waves**. That is precisely **matter: confined waves**.

Thanks to Max Planck's work, the **mass of particles** allows us to determine their wavelength λ, that is, their frequency. Therefore, any confinement must satisfy:

$$L = n\lambda, \quad n \in \mathbb{Z}^+$$

If we know λ, it is equivalent to knowing the **length of the perimeter** where the waves are confined. From this, we can calculate the **associated radius** of the confinement. Although the contour may have a complex shape, it is still a calculable geometric condition of the universe.

In fact, this **geometric and deterministic** idea allows us to calculate with precision the **radius of the proton**. In a 4-dimensional space, we observe that at least **n=4** is required for the wave to confine correctly, indicating that **n has a direct relationship with the dimensionality and volume of the confinement space**.

This allows us to obtain the proton radius in a **non-probabilistic, exact, and deterministic** way, with a **precision much greater than any current experimental measurement**, posing a **future verification challenge**.

$$R_p = \frac{n\hbar}{m_p c} = \frac{4 \cdot 1.0545718 \times 10^{-34}\,\mathrm{J{\cdot}s}}{(1.6726 \times 10^{-27}\,\mathrm{kg})(2.9979 \times 10^{8}\,\mathrm{m/s})} \approx 8.42 \times 10^{-16}\,\mathrm{m}$$

Comparing with **current experimental limits**, which place the proton radius approximately at:

$$R_p^{\mathrm{exp}} \sim 0.841 \times 10^{-15}\,\mathrm{m} \pm 0.001\,\mathrm{fm}$$

> The coincidence between the calculated value and the experimental ranges demonstrates that the **underlying geometry of wave confinement is the key that unites the very large with the very small**, providing a direct bridge between elementary particles and cosmological scales.

Therefore, we can state rigorously:

> **Matter is confined waves.**

Thus we define: - **Particles and energy**: as modes of spacetime, where energy is free and matter is confined. - **Content and container**: the quality of the content (waves) and the geometry that limits them (confinement).

Type	Dimension n	Example	Geometry
3D volumetric	3	Proton	Volume
2D surface	2	Electron	Surface
2D limit	2	Black hole horizon	Saturated surface

2.4 Einstein-Planck unification

From Einstein: $E = mc^2$ From Planck: $E = h\nu = \frac{hc}{\lambda}$

Equating both expressions for energy:

$$mc^2 = \frac{hc}{\lambda} \quad \Rightarrow \quad \lambda = \frac{h}{mc}$$

This is the **Compton wavelength** reinterpreted as the natural wavelength of the spacetime vibration associated with mass m.

2.5 Geometric condition of confinement

For circular confinement:

$$2\pi R = n\lambda = n \cdot \frac{h}{mc}$$

Simplifying with $\hbar = h/(2\pi)$ we obtain the **fundamental equation of the VED radius**:

$$\boxed{R = \frac{n\hbar}{mc}}$$

or equivalently:

$$\boxed{m = \frac{n\hbar}{Rc}}$$

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3. VED RADIUS AND MASS

3.1 Definition of VED radius

$$R_{\text{VED}} = \frac{n\hbar}{mc}$$

3.2 Mass as an emergent phenomenon

Mass is not a primitive property, but **emerges** from: - n: number of confinement modes (integer) - R: confinement radius (length)

The relation $m \propto n/R$ shows that for a fixed radius, more modes (larger n) imply more mass, and for a fixed n, a larger radius implies less mass.

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4. RELATIONSHIP WITH GRAVITY

4.1 Schwarzschild radius

From the Schwarzschild solution in General Relativity:

$$R_s = \frac{2Gm}{c^2}$$

4.2 Relationship between VED and Schwarzschild radii

For the **same physical mass** m, we equate both expressions:

$$\frac{n\hbar}{R_{\text{VED}}c} = \frac{R_s c^2}{2G}$$

Rearranging we obtain the **fundamental geometric relation**:

$$\boxed{R_{\text{VED}} \cdot R_s = \frac{2n\hbar G}{c^3}}$$

4.3 The gravitational constant as an emergent quantity

Solving for G from the previous relation:

$$G = \frac{R_{\text{VED}} \cdot R_s \cdot c^3}{2n\hbar}$$

In natural units ($c = \hbar = 1$), this simplifies to:

$$G = \frac{R_{\text{VED}} \cdot R_s}{2n}$$

Which shows that G is essentially an **"area per mode" divided by 2**.

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5. EMPIRICAL PROOF: THE QUANTIZATION OF G

5.1 The predictive formula $G(n)$

Substituting the explicit expressions of R_{VED} and R_s into the formula for G, we obtain the **quantized prediction**:

$$G(n) = G_{\text{exp}} \cdot \frac{2}{n}$$

where $G_{\text{exp}} = 6.67430 \times 10^{-11}\,\mathrm{m^3/kg{\cdot}s^2}$ is the experimentally measured value.

5.2 Experimental verification: the "target" of G

This formula makes a **verifiable prediction**: for each integer n, G takes a different discrete value.

5.3 Exact numerical precision for n=2

Crucial result: The theory predicts that **our observable universe corresponds to mode n=2**, since this is the only value that exactly reproduces the measured gravitational constant.

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6. SPECTRUM OF n AND PARTICLES

6.1 n as fundamental geometric quantum number

In VED, n is not a postulated quantum number, but **completely determines** the properties of each system: - Small $n \to$ strong confinement, small radius for given mass - Large $n \to$ weak confinement, large radius for given mass

6.2 Specific values for known particles

For the proton: With $m_p = 1.6726 \times 10^{-27}\,\mathrm{kg}$ and $R_p = 0.841 \times 10^{-15}\,\mathrm{m}$:

$$n_p = \frac{R_p m_p c}{\hbar} \approx 4.00$$

For the electron in the hydrogen atom: With $m_e = 9.109 \times 10^{-31}\,\mathrm{kg}$ and $R_{\text{Bohr}} = 5.29 \times 10^{-11}\,\mathrm{m}$:

$$n_e = \frac{R_{\text{Bohr}} m_e c}{\hbar} \approx 137.04 \approx \alpha^{-1}$$

where $\alpha \approx 1/137$ is the fine structure constant.

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7. CONFINEMENT VS FREE ENERGY

7.1 The limit n → ∞

When $n \to \infty$, the confinement radius becomes:

$$\lim_{n\to\infty} R_{\text{VED}} = \lim_{n\to\infty} \frac{n\hbar}{mc} \to \infty$$

This corresponds to **free waves** (photons) that are not confined and therefore do not generate gravitational effects.

7.2 Energy as the unconfined limit

Energy in VED is simply the limit case where waves are not confined within a finite region. This explains why: - Photons have zero rest mass - Gravitational lensing affects photons not because they have mass, but because they follow the curvature of spacetime - Electromagnetic waves do not create static gravitational fields

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8. CONTAINER AND CONTENT

8.1 The holographic principle in VED

The universe can be understood as: - **Container**: 3S+T spacetime continuum - **Content**: confined waves (matter) and free waves (energy)

8.2 Emergence of 3D reality from 2D information

VED suggests that all physical information is encoded on a 2D surface (the holographic principle), and the 3D reality we perceive emerges from the interaction of these surface modes.

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9. DETERMINISM AND REALISM

9.1 Critique of Copenhagen interpretation

VED provides an alternative to the probabilistic interpretation of quantum mechanics: - Wave function describes real waves in spacetime - Measurement reveals pre-existing properties - No "collapse" of the wave function

9.2 Hidden variables reinterpreted

In VED, the "hidden variables" are simply the geometric parameters (n, R, confinement shape) that completely determine the system's behavior.

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10. CONCLUSIONS AND PREDICTIONS

10.1 Summary of achievements

1. Derived proton radius from first principles with n=4
2. Explained fine structure constant as geometric consequence
3. Predicted and verified quantization of G with n=2

4. Unified matter, energy, and gravity in geometric framework
5. Provided deterministic alternative to quantum probability

10.2 Testable predictions

1. **Proton radius**: $R_p = 0.841 \times 10^{-15}$ m (testable with improved measurements)
2. **G quantization**: $G(n) = G_{\text{exp}} \cdot 2/n$ (testable with precision G measurements)
3. **Electron orbitals**: Successive orbitals correspond to n = 137, 138, 139...
4. **New particles**: Should correspond to integer n values

10.3 Philosophical implications

- Reality is fundamentally geometric and deterministic
- Physical constants emerge from geometry, not as fundamental parameters
- The holographic principle provides a unifying framework
- Quantum mechanics can be reformulated as deterministic wave theory

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References and further reading: - Original theory development: estradad.es/teorias - Mathematical derivations: See complete PDF documentation - Experimental verification: Precision measurements of G and proton radius